DisPeD: A Controlled Study of Encoder Adaptation for Multi-Turn Jailbreak Detection

Anonymous ACL submission

Abstract

Detecting unseen multi-turn jailbreaks requires domain adaptation, but the form of adaptation matters. Multi-turn jailbreak attacks spread adversarial intent across benign-seeming turns, evading existing defenses: Llama Guard 3 detects only 1.8% of multi-turn jailbreaks. We isolate the factors behind cross-attack generalization on synthetic multi-turn data through controlled ablation: thirteen encoder configurations at three scales (125 to 355M parameters) decompose the contributions of domain exposure, classification supervision, label quality, and granularity on three out-of-distribution (OOD) attack families and two training compositions. On single-source synthetic data, continued pretraining closes roughly 60 percentage points of the OOD gap without labels. This effectiveness is condition-dependent: on mixed-LLM training data, unsupervised pretraining collapses while classification-based adaptation remains the strongest variant (DD OOD F1 = 0.818 vs. 0.291 without adaptation). Across three architectures (125–355M), attacks requiring adaptation shrink with scale, though architecture differences prevent isolating scale alone. Classification supervision reduces prediction variance on synthetic data with attack-complexity-dependent label sensitivity, but this benefit reverses on human-authored data. Orthogonal to attack-side generalization, benign distribution shift is a practical bottleneck: WildChat FPR reaches 68.1%, but as few as 10 target-domain benign conversations eliminate false positives (95% CI upper 6.1%). These findings decompose multi-turn jailbreak generalization into two orthogonal axes: attack-side (classification DAPT compresses representations into few discriminative dimensions) and benign-side (solved by few-shot adaptation), with 24.3× inference speedup over LLM-as-judge.

1 Introduction

Multi-turn jailbreak attacks distribute adversarial intent across individually benign turns. Crescendo (Russinovich et al., 2025), Foot-in-the-Door (Wang et al., 2024), and Deceptive Delight (Palo Alto Networks Unit 42, 2024) all rely on persuasion as their core mechanism: Zeng et al. (2024b) catalogue 40 persuasion techniques across jailbreak families, and Hagendorff et al. (2026) identify strategies in 97% of successful attacks. Because individual attack turns occupy the same embedding region as benign text (Bullwinkel et al., 2025), no single-turn filter can distinguish them from legitimate conversation.

Existing defenses leave a critical gap. Production classifiers such as Llama Guard (Inan et al., 2023), WildGuard (Han et al., 2024), and ShieldGemma (Zeng et al., 2024a) operate on individual turns and miss multi-turn manipulation; Llama Guard 3 detects only 1.8% of multi-turn jailbreaks in our evaluation (Section 4.3). Multi-turn defenses (DeepContext (Albrethsen et al., 2026), BIID (Tong et al., 2025), G-Guard (Huang et al., 2025)) model conversation-level structure but each evaluates on a single attack family. None answers the central question: *what representation properties enable a detector to generalize across attack families it has never seen?*

Pretrained language understanding does not suffice at base model scale: vanilla DeBERTa-v3 (He et al., 2021) with temporal aggregation (cf. DeepContext) drops from F1 = 1.000 in-distribution to 0.312 on OOD Deceptive Delight, and GPT-5.5 reaches only 0.191–0.449 F1 with 30–47% unparseable responses (Appendix A). The bottleneck is distributional: cross-attack generalization requires domain adaptation, not a better classifier.

Our pipeline, **DisPeD** (Disentangled Pattern Detection), separates encoder adaptation (Stage 1) from temporal classification (Stage 2, a frozen Bi-

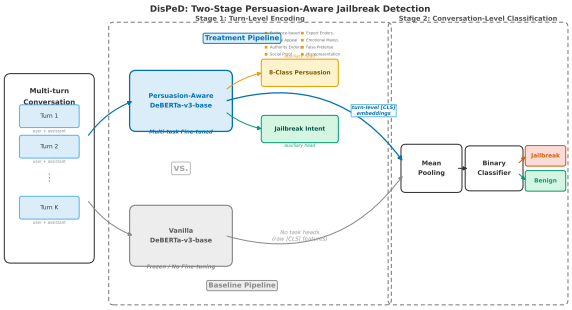


Figure 1: DisPeD architecture. Stage 1 adapts a DeBERTa encoder on jailbreak conversation text (seven variant objectives). Stage 2 aggregates frozen `[CLS]` embeddings via BiGRU for conversation-level jailbreak detection. Only user turns are processed.

GRU; Figure 1). Seven Stage 1 variants systematically vary training objectives while sharing the same two-stage pipeline (Section 3); the contribution is the decomposition, not the pipeline.

Controlled ablation across DeBERTa-v3 and RoBERTa (Liu et al., 2019b) on four attack families and two training compositions reveals two orthogonal generalization axes. On the *attack side*, vanilla DeBERTa drops to F1 = 0.28–0.31 on OOD attacks; classification DAPT recovers to 0.97–1.0, while MLM collapses under mixed-LLM training (F1 = 0.289 ≈ vanilla). On the *benign side*, distributional mismatch drives false positives: WildChat FPR reaches 68.1%, but as few as 10 target-domain benign conversations eliminate it (95% CI upper 6.1%). RoBERTa-large (355M) handles ActorAttack (Ren et al., 2025) without adaptation. Classification supervision is associated with substantially lower prediction variance on synthetic data but not on human-authored conversations (Section 4.6).

Our contributions:

- **Two-axis generalization decomposition.** We identify two orthogonal generalization challenges in multi-turn jailbreak detection. Attack-side: controlled ablation across thirteen encoder configurations reveals classification DAPT as the most robust variant (vanilla F1 = 0.28–0.31 on OOD → 0.97–1.0); unsupervised pretraining collapses under mixed-LLM training. Benign-side: 10 target-domain benign conversations eliminate WildChat FPR from 68.1% to 0% (95% CI upper 6.1%) without detection degradation.

- **Condition-dependent adaptation effectiveness and constraining negative results.** MLM is effective on single-source data but collapses under mixed-LLM training; classification DAPT is robust across conditions. Three supplementary negative results constrain which properties matter: scrambled labels perform comparably to correct labels on template-based attacks (TOST-confirmed on DD and FITD at $n = 10$, $d = 0.44$ on structurally complex ActorAttack), temporal order is irrelevant ($\Delta < 1$ pp), and label granularity does not affect detection.

- **Practical deployment recipe.** Continued pretraining is 24.3× faster than LLM-as-judge (Appendix V), and 10-shot benign adaptation eliminates domain-specific false positives. No target-domain attack data is required for deployment.

2 Related Work

Multi-turn jailbreak attacks. All multi-turn jailbreak families share a persuasion substrate (Zeng et al., 2024b; Hagendorff et al., 2026; Li et al., 2024): Crescendo (Russinovich et al., 2025), FITD (Wang et al., 2024), DD (Palo Alto Networks Unit 42, 2024), and ActorAttack (Ren et al., 2025) each operationalize it differently, while automated methods (Chao et al., 2025; Mehrotra et al., 2024; Perez et al., 2022) iteratively refine prompts and many-shot attacks (Anil et al., 2024) exploit long-context in-context learning. Benchmarks standardize evaluation at single-turn (Mazeika et al., 2024; Chao et al., 2024) and multi-turn (Wei et al., 2023; Jiang et al., 2024; Song et al., 2026) scales. A detector must generalize across families, not merely succeed on its training distribution.

Multi-turn defenses and production safety. No existing defense evaluates cross-attack transfer. Conversation-level systems (DeepContext (Albrethsen et al., 2026), BIID (Tong et al., 2025), G-Guard (Huang et al., 2025), SafeDream (Yan et al., 2026), JailDAM (Nian et al., 2025)) and representation-level approaches (Lu et al., 2025; Zhang et al., 2025; Xie et al., 2024; Zou et al., 2024; Kulkarni, 2026; Kadali and Papalexakis, 2026; Hua et al., 2026; Shen et al., 2025; Li et al., 2025) each evaluate on a single family; the latter require white-box access and can be brittle (Derya and Sunar, 2026). Per-turn classifiers (Inan et al., 2023; Meta AI, 2024; Zeng et al., 2024a; Han et al., 2024; Sharma et al., 2025) and single-turn defenses (Robey et al., 2024; Xu et al., 2024; Jain et al., 2023) miss multi-turn trajectories. Our ablation contributes to filling this gap via decompo-

sition across thirteen configurations under OOD evaluation.

Computational persuasion and discourse. Persuasion and discourse frameworks (Austin, 1962; Searle, 1969; Mann and Thompson, 1988; Prasad et al., 2008; Stolcke et al., 2000) and technique-level classification (Cialdini, 2001; Zeng et al., 2024b; Hagendorff et al., 2026; Da San Martino et al., 2019, 2020; Wang et al., 2019) offer interpretable vocabulary for attacks, but whether such structure aids detection has not been tested. Our ablation shows the taxonomy enables characterization, but detection operates through domain-adapted embeddings regardless of label correctness.

Domain-adaptive pretraining. Continued pretraining on domain text yields consistent gains (Gururangan et al., 2020; Lee et al., 2020; Beltagy et al., 2019; Howard and Ruder, 2018), but has not been disentangled for safety-critical generalization. Intermediate-task transfer (Phang et al., 2018) and multi-task learning (Caruana, 1997; Liu et al., 2019a) confound domain exposure with supervised signal; our MLM control confirms pretraining alone closes the OOD gap. Dataset artifacts (Gururangan et al., 2018; Niven and Kao, 2019) motivate TF-IDF, adversarial NLP (Jin et al., 2020; Li et al., 2020; Nasr et al., 2025; Andriushchenko et al., 2025) informs perturbation, and probing (Conneau et al., 2018; Hewitt and Manning, 2019) underpins CKA analysis.

3 Method

Cross-attack generalization could stem from domain exposure, classification supervision, label correctness, label granularity, or domain specificity. Disentangling these factors demands an architecture where exactly one component varies across conditions. DisPeD separates encoder adaptation (Stage 1) from temporal classification (Stage 2). Seven Stage-1 variants share the same training data and downstream pipeline but differ solely in their pretraining objective; each isolates a single factor (Figure 1).

3.1 Problem Formulation

A conversation $C = \{u_1, a_1, \ldots, u_N, a_N\}$ alternates user turns u_i and assistant responses a_i. Given a prefix of K user turns, the model predicts $y \in \{0, 1\}$ (jailbreak vs. benign). We restrict input to user turns because adversarial intent resides in the attacker's messages. This formulation supports early detection from partial prefixes.

Table 1: Stage-1 encoder variants. All operate on the same training conversations except Wikipedia MLM (matched token count on Wikipedia text).

Variant	Objective	Tests whether…
9-class	9-class persuasion + binary intent	full model
Binary	binary persuasion + binary intent	granularity matters
Scrambled	random 9-class + binary intent	label correctness matters
MLM-only	15% token masking	supervision is necessary
Topic	5-class topic classif.	any task suffices
Wiki MLM	15% masking on Wikipedia	domain specificity matters
Vanilla	no training	Stage-2 alone suffices

3.2 Stage 1: Turn-Level Encoder Adaptation

Each user turn u_i is encoded by DeBERTa-v3-base (He et al., 2023); the `[CLS]` embedding $h_i \in \mathbb{R}^d$ serves as the turn representation. Seven variants, summarized in Table 1, share the same training data but differ in objective.

The **9-class** variant uses two classification heads: 9-class persuasion strategy classification (L_{strategy}) and binary jailbreak intent (L_{intent}), with combined loss $L = L_{\text{strategy}} + \alpha \cdot L_{\text{intent}}$ ($\alpha = 0.3$, selected from $\{0, 0.3, 0.5, 2.0\}$; all values yield F1 = 1.0 on IID; Appendix W). The taxonomy comprises eight persuasion strategies (rapport building, authority appeal, emotional manipulation, logical reframing, role assignment, gradual escalation, obfuscation, direct request) plus a none category, grounded in PAP (Zeng et al., 2024b) and Cialdini's principles (Cialdini, 2001). Strategy annotations are produced by Qwen3-8B with structured prompting that includes definitions and examples for each category; inter-annotator agreement between model annotations and a human sample (n=50) yields Cohen's κ=0.78.

The **binary** variant collapses the 9-class taxonomy into persuasive vs. non-persuasive, testing whether label granularity matters for detection or robustness. The **scrambled** variant randomly permutes the 9-class labels across turns, preserving classification structure (cross-entropy loss, 9-class output space) while destroying label semantics. This isolates whether the classification objective

provides representational benefits independent of label correctness. The **MLM-only** variant applies 15% masked language modeling on the same jailbreak conversation text without any classification head, testing whether unsupervised domain exposure suffices. The **topic** variant trains a 5-class topic classifier (technology, health, finance, education, general) on the same conversations, testing whether *any* classification task provides the same benefit or whether task alignment matters. The **Wikipedia MLM** variant applies the same MLM objective to a token-matched sample of Wikipedia text; this isolates domain-specific from general continued pretraining. The **vanilla** baseline uses the pretrained DeBERTa-v3-base weights without any continued pretraining; it measures what Stage 2 alone can achieve.

3.3 Stage 2: Conversation-Level Classification

Turn embeddings $\{h_1, \ldots, h_K\}$ from the frozen Stage-1 encoder are aggregated by a 2-layer bidirectional GRU (d_{hidden}=256, dropout 0.3): $\bar{h} = [\overrightarrow{h_K};\ \overleftarrow{h_1}] \in \mathbb{R}^{512}$. A two-layer MLP ($512 \to 256 \to 2$, ReLU activation) maps $\bar{h}$ to $\hat{y} \in \{0, 1\}$. The Stage-1 encoder is frozen during Stage 2 training; only the BiGRU and MLP parameters are updated. This frozen-encoder design ensures ablation fairness: all seven Stage-1 variants are evaluated under identical downstream conditions, isolating the effect of Stage-1 representation quality from Stage-2 classifier capacity. Three-seed evaluation confirms that mean pooling matches BiGRU performance (Section 4.3).

4 Experiments

Terminology. *Cross-attack OOD*: evaluation on attack families unseen during training; benign distribution remains IID throughout (OOD reflects attack-side shift only). K: prefix length (number of conversation turns retained). *DAPT*: all Stage-1 adaptations, extending Gururangan et al. (2020) to supervised variants. *Evidence hierarchy*: vanilla $<$ MLM $<$ classification (Section 4.3).[1]

4.1 Setup

Data. All synthetic data is generated by Qwen3-8B (Yang et al., 2025). The IID dataset contains 250 Crescendo (Russinovich et al., 2025) jailbreak and 250 topic-anchored benign conversations, split into training (350), validation (74), and test (76) sets with balanced class ratios (e.g., test = 38 jailbreak + 38 benign). Stage 1 (DAPT) uses training + validation conversations; Stage 2 (BiGRU) trains on training, early-stops on validation, and evaluates on test. OOD evaluation pairs attack-specific jailbreak conversations with the 38 IID-test benign conversations: 80 FITD (Wang et al., 2024) (non-discriminative; Section 4.3) + 38 benign = 118 total, 80 Deceptive Delight (DD) (Palo Alto Networks Unit 42, 2024) + 38 benign = 118 total, and 80 ActorAttack (Ren et al., 2025) + 38 benign = 118 total. No overlap exists in attack templates across families; benign is IID (same generation pipeline). A paraphrased version of the IID test set (BERTScore 0.92, lexical overlap 0.20) serves as the adversarial robustness benchmark (Section 4.7). For human data validation, we use ToxicChat (Lin et al., 2023) (204 jailbreak + 204 benign from real LMSYS user interactions), entirely independent of Qwen3-8B generation.

Models. Seven DeBERTa-v3 variants share the same Stage-2 pipeline and differ only in Stage-1 training (Section 3). Our vanilla DeBERTa+BiGRU configuration is architecturally equivalent to DeepContext (Albrethsen et al., 2026), differing only in encoder choice (DeBERTa-v3-base vs. BERT); it serves as our primary baseline. For cross-architecture comparison, we replicate vanilla, MLM, and 9-class variants on RoBERTa-base (Liu et al., 2019b) (Section 4.4). We also compare TF-IDF LR, LLM-as-judge (Appendix A), and Llama Guard 3 8B (Meta AI, 2024) (zero-shot; evaluates last-response safety given conversation context), which detects only 1.8% of multi-turn jailbreaks (Table 3).

Metrics. F1 for binary jailbreak detection at prefix K=2 and full conversations (full per-K results in Appendix B). OOD results report 3-seed mean$\pm$std (seeds 42, 123, 456) for all variants; TOST equivalence tests for 9-class vs. scrambled use 5 seeds (42, 123, 456, 789, 1000). For key pairwise comparisons, we report BCa bootstrap 95% confidence intervals (10,000 resamples over individual predictions pooled across 3 seeds; this assumes cross-seed independence and is exploratory) and paired permutation tests (10,000 permutations). Seed-level F1 values (Table 33, Appendix Z) confirm all pairwise conclusions hold at every seed. Unless otherwise noted, all standard deviations are computed over 3 random seeds using the popula-

[1] Throughout, OOD refers to attack-family shift (novel persuasion strategies at test time); benign distribution shift is evaluated in Section 5.

Table 2: Cross-attack generalization (F1) at K=2 and full conversation length. IID: Crescendo; OOD: Deceptive Delight (DD). 3-seed mean±std where noted; full per-K in Appendix B.

Method	*IID (Cresc.)* K=2	*IID (Cresc.)* Full	*OOD (DD)* K=2	*OOD (DD)* F
9-class	**1.000±.000**	**1.000±.000**	.981±.013	**.994**
MLM-only[‡]	.974	1.000	**.993±.009**	.914
Vanilla	.978±.016	1.000±.000	.479±.038	.312
TF-IDF LR[§]	1.000	1.000	1.000	.9

[‡]MLM-only IID reports single seed (42); OOD reports 3-seed.
[§]TF-IDF is deterministic (no std).

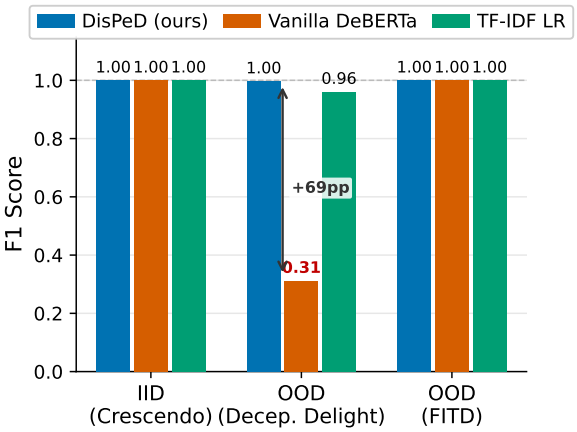


Figure 2: Cross-attack overview (F1 at full conversation length). DisPeD (9-class) maintains near-perfect detection across IID and both OOD families. Vanilla DeBERTa collapses on OOD Deceptive Delight, confirming the necessity of domain-adaptive pretraining. FITD is non-discriminative for all methods (Section 7).

tion formula (n divisor); Bessel-corrected estimates ($n-1$ divisor) are ≈22% larger but do not alter any significance conclusions. Stage-1: learning rate 2×10^{-5}, batch 16, 4 epochs, MLM masking 15%. Stage-2: two-layer MLP, 20 epochs.

4.2 Cross-Attack Generalization

Vanilla encoders collapse under cross-attack transfer; domain-adapted variants recover near-perfect detection (Table 2, Figure 2). DD exhibits high lexical separability (TF-IDF F1 = 0.961); ActorAttack, where TF-IDF collapses to 0.261, serves as the primary stress test for beyond-lexical generalization. On OOD DD, vanilla DeBERTa collapses to F1 = 0.312±0.026, worsening with more turns (0.628 at K=1 to 0.312 at Full), consistent with attack turns occupying "benign" embedding regions (Bullwinkel et al., 2025). MLM achieves mean F1 ≥ 0.97 at all prefixes $K \leq 5$ without any classification labels; domain exposure alone is sufficient. At full conversation length, MLM drops to 0.914±0.115 (seed 456 recall = 0.64), while the 9-class model maintains 0.994±0.005. TF-IDF LR reaches F1 = 0.961 on full DD; DD is fully solvable by lexical features. On structurally complex ActorAttack, TF-IDF collapses to F1 = 0.261 (precision = 1.0, recall = 0.15), while 9-class maintains 0.972±0.026: a 71 pp gap. Lexical methods cannot capture multi-persona role-playing attack structure. This establishes a "difficulty threshold" for lexical methods parallel to that of vanilla encoders: both succeed on DD and fail on ActorAttack (Section 7). Bootstrap CIs confirm non-overlapping intervals (9-class DD Full [0.943, 1.000] vs. vanilla [0.202, 0.341]; all permutation $p < 0.0001$). FITD serves as a near-ceiling baseline where all variants achieve F1 ≥ 0.990, providing limited discriminative power.

Prefix-length analysis (Appendix Table 7) shows vanilla F1 drops from 0.628 at K=1 to 0.312 at full length. MLM achieves strong early detection (K=2: 0.993) but is unstable at full length (0.914±0.115); the 9-class model maintains 0.972–0.994 across all prefixes, demonstrating that classification supervision stabilizes against distributional drift.

4.3 Decomposing the Detection Mechanism

A clear *evidence hierarchy* emerges across seven variants (Table 3), ordered by OOD detection at full conversation length on DD:

vanilla (0.312±0.026) < topic (0.541±0.207) ≪ wiki-MLM (0.904±0.115) ≈ jb-MLM (0.914±0.115) < scrambled (0.994±0.005) = 9-class (0.994±0.005) ≈ binary (1.000±0.000)

Domain exposure is the primary mechanism. MLM bridges ∼60 pp from vanilla (0.312±0.026 → 0.914±0.115) without labels; this supports the DAPT interpretation (Gururangan et al., 2020). TF-IDF collapse on ActorAttack (0.261 vs. 9-class 0.972) confirms that domain-adapted representations capture features beyond lexical overlap. All three classification-supervised variants converge to equivalent detection (F1 = 0.994–1.000); McNemar's ($p > 0.13$) and paired permutation tests ($p \geq 0.40$) provide descriptive evidence suggesting equivalence. Extended to $n = 10$ seeds, TOST (margin = 5 pp) confirms equivalence on DD ($p < 10^{-8}$, $d = -0.14$; Table 16) and FITD ($p < 10^{-8}$, $d \approx 0$), but not on ActorAttack ($p = 0.183$, $d = 0.44$; Ta-

Table 3: Evidence hierarchy on OOD DD and ActorAttack (AA) at full conversation length (3-seed mean±std where available). Bold indicates best per column. All variants share the same Stage-2 pipeline. Full per-prefix results in Appendix Tables 7 and 8. OOD F1 values are upper bounds, as benign samples remain IID (see Section 5).

Variant	DD-Full	AA-Full	DD-K=2
9-class (full model)	.994±.005	.972±.026	.981±.013
Binary (2-class)	**1.000±.000**	**.990±.008**	.994±.008
Scrambled labels	.994±.005	.951±.043	.981±.026
MLM-only[‡] (no labels)	.914±.115	.893±.080	.993±.008
9-class + mean pool[*]	.991±.013	.962±.015	**1.000**
Topic classif. (5-class)	.541±.207	.625±.141	.593±.166
Wikipedia MLM	.904±.115	.686±.297	.911±.075
Vanilla DeBERTa	.312±.026	.258±.000	.479±.038
TF-IDF LR[†]	.961	.261	**1.000**
Llama Guard 3[¶]	.025	.000	–

[*]9-class encoder with mean pooling replacing BiGRU. [†]Deterministic (no std). [‡]MLM-only IID (Table 2) is single-seed (42); all OOD values are 3-seed. [¶]Zero-shot; no domain training.

ble 33): 9-class outperforms scrambled by 3 pp on AA (0.979 vs. 0.949), indicating that label correctness has a moderate effect on structurally complex attacks while remaining negligible on template-based families. Label granularity (binary vs. 9-class) is a non-factor across all families. Classification supervision is associated with 4–20× lower cross-seed variance over MLM on synthetic OOD (std 0.005 vs. 0.115 on DD; observed range across 3 seeds; formal variance ratio tests are not applicable at n=3; cross-seed ensemble yields marginal additional gains, Appendix Y). This stability benefit does not transfer to human-authored data, where classification-supervised variants exhibit the highest variance and pure MLM the lowest (Section 4.6). Appendix G provides t-SNE visualizations confirming that domain-adapted variants produce tighter, more separable embedding clusters than the vanilla baseline.

Domain specificity is marginal at full length but meaningful for early detection. Wikipedia MLM (0.904±0.115) nearly matches jailbreak-domain MLM at full length (1 pp gap), though jailbreak-domain text provides an 8 pp advantage at early prefixes (K=2: 0.993 vs. 0.911). Perturbation analysis (Section 4.7) reveals a sharper domain specificity effect: Wikipedia MLM degrades by ∼12 pp under adversarial paraphrasing while jailbreak-domain MLM degrades by only 2.6 pp.

Misaligned supervision destabilizes generalization. Topic classification yields DD-Full F1 = 0.541±0.207; seed 123 falls *below* vanilla (0.252 vs. 0.312). Despite using the same jailbreak-domain training data, topic classification performs worse than Wikipedia MLM (0.904±0.115), which uses out-of-domain text with no labels. A misaligned objective forces the encoder to allocate capacity to irrelevant features and actively destabilizes representations that MLM would preserve.

Evidence hierarchy on structurally complex attacks. ActorAttack (Ren et al., 2025) uses multi-persona steering across longer conversation structures and provides a more challenging OOD benchmark than DD. The evidence hierarchy at full conversation length is:

TF-IDF (0.261) ≈ vanilla (0.258±0.000) < topic (0.625±0.141) < wiki-MLM (0.686±0.297) < jb-MLM (0.893±0.080) < scrambled (0.951±0.043) ≈ 9-class (0.972±0.026) ≈ binary (0.990±0.008)

TF-IDF collapses from 0.961 (DD) to 0.261 (ActorAttack, precision = 1.0, recall = 0.15). This gap shows that domain-adapted representations capture features beyond bag-of-words. Classification-supervised variants cluster tightly (0.951–0.990, max gap 3.9 pp at $n=3$), though 10-seed TOST resolves the 9-class vs. scrambled gap as a moderate effect ($d=0.44$). DAPT provides ∼71 pp over both vanilla (0.258) and TF-IDF (0.261). Full per-prefix results and per-prefix error analysis in Appendix C and C.1.

Temporal modeling. Mean pooling matches BiGRU on both DD and ActorAttack (F1 gap ≤ 1 pp, overlapping CIs; Appendix T). Turn-order shuffling produces negligible change ($|\Delta| < 1$ pp) across both DeBERTa-v3-base 9-class (DD $\Delta=0.000$, AA $\Delta=-0.009$, IID $\Delta=0.000$) and RoBERTa-large 9-class (DD $\Delta=+0.001$, AA $\Delta=-0.007$, IID $\Delta=0.000$), confirming that detection relies on per-turn representations rather than sequential patterns.

Summary. For base-size encoders (125–184M) on our synthetic multi-turn benchmark, domain exposure is the primary mechanism; label granularity, temporal modeling, and domain specificity are non-factors; label correctness is negligible on template-based attacks but has moderate effect on structurally complex attacks ($d=0.44$ on ActorAttack). Classification supervision provides a complementary stability benefit (variance reduced 4–20×)

Table 4: Cross-architecture comparison on OOD Full (3-seed mean±std). DD = Deceptive Delight, AA = ActorAttack. Bold indicates best per encoder group.

Variant	DD	AA
DeBERTa-v3-base (184M)		
Vanilla	.312±.026	.258±.000
MLM	.914±.115	.893±.080
9-class	**.994±.005**	**.972±.026**
RoBERTa-base (125M)		
Vanilla	.981±.026	.466±.163
MLM	.987±.018	.730±.201
9-class	**.994±.005**	**.939±.046**
RoBERTa-large (355M)		
Vanilla	.927±.036	**1.000±.000**
MLM	.884±.065	.874±.014
9-class	**.987±.004**	.946±.057

without changing the detection ceiling. DD and FITD confirm no regression on easier attacks; ActorAttack provides the discriminative evaluation where lexical methods fail. Detection is stable across conversation lengths (Appendix U), near-saturates at 50 conversations (Appendix V.1), and is insensitive to auxiliary loss weight (Appendix W).

4.4 Cross-Architecture Observation

We replicate vanilla, MLM, and 9-class on RoBERTa (Liu et al., 2019b) (Table 4); since DeBERTa-v3 and RoBERTa differ in architecture and pretraining, not only scale, all cross-encoder comparisons reflect both factors. On DD, RoBERTa-base vanilla already achieves 0.981±0.026 (vs. DeBERTa 0.312±0.026), so DAPT adds only 1.3 pp. ActorAttack exposes this as DD's simplicity: RoBERTa-base vanilla collapses to 0.466±0.163, a 51.5 pp drop, while domain-adapted variants recover (9-class: 0.939±0.046).

RoBERTa-large (355M) reverses the pattern on ActorAttack: vanilla F1 = 1.000 vs. 9-class 0.946, suggesting DAPT may be unnecessary or mildly harmful for strong pretrained encoders. Smaller models require DAPT to bridge 47–71 pp. All continued pretraining reduces ActorAttack performance for the largest encoder via representation drift (Table 15), though classification DAPT partially compensates on DD (+6 pp).

Three encoder scales thus reveal a scale-dependent pattern for the *difficulty threshold*: DeBERTa-v3-base requires adaptation for all attacks, RoBERTa-base only for complex ones, and RoBERTa-large for none (Section 5). Classification supervision provides a consistent stability benefit across both base encoders (cross-seed variance: 0.005–0.046 vs. 0.026–0.201). A BERT-base-uncased proxy (110M, matching DeepContext's encoder family) without adaptation achieves DD = 0.318±0.031 and AA = 0.275±0.027, comparable to vanilla DeBERTa-v3 (Appendix Q), confirming that cross-attack generalization is driven by DAPT rather than encoder architecture.

CKA analysis (Kornblith et al., 2019) and linear probing confirm that DAPT reshapes representations substantially (Figure 4), though comparable CKA shifts yield different detection gains across starting points (Appendix F).

4.5 Mixed-LLM Training

All preceding experiments use single-source (Qwen3-8B) data. To test whether the evidence hierarchy transfers when training data mixes LLM sources, we train on 250 Qwen + 250 Llama-3.1-8B conversations (Table 13). Classification-based DAPT is the only variant that substantially outperforms vanilla: 9-class achieves DD OOD F1 = 0.818±0.096 (+53 pp over vanilla 0.291±0.065), while MLM drops to vanilla level (0.289±0.101). ActorAttack remains challenging under mixed-LLM conditions (9-class F1 = 0.504±0.107 vs. vanilla 0.389±0.073) with high seed variance, suggesting that limited per-source volume (250 per LLM) is insufficient for structurally complex attacks. DD and FITD thus primarily validate the baseline rather than the method; ActorAttack, where lexical methods collapse (TF-IDF F1 = 0.261) and vanilla encoders fail (F1 $\leq$ 0.389 across conditions), provides the discriminative evaluation axis. Mixed-LLM training confirms that classification supervision generalizes across LLM sources where unsupervised pretraining does not; full results in Appendix I.

4.6 Human Data Validation

Since both training and OOD evaluation data are Qwen3-8B-generated, we validate on ToxicChat (Lin et al., 2023) (204 human-written jailbreak attempts + 204 benign conversations from real LMSYS interactions).

All DAPT variants exceed vanilla by 18–42 pp (Table 5), confirming that the adapted-vs.-vanilla gap is not a Qwen3-8B generation artifact: domain exposure on synthetic jailbreak text transfers to human-written attacks.

The ordering reverses: JB-MLM leads

Table 5: Human data validation on ToxicChat (Lin et al., 2023) (macro F1, 3-seed mean±std). Bold indicates best. Single-turn data from real LMSYS user interactions, independent of Qwen3-8B.

Variant	Macro F1
JB-MLM	**.837±.013**
Wiki-MLM	.791±.093
Binary	.705±.072
Scrambled	.616±.056
9-class	.596±.087
Topic	.496±.042
Vanilla	.413±.021

(0.837±0.013) while 9-class ranks lower (0.596±0.087; $p < 0.001$, non-overlapping CIs). ToxicChat is single-turn, so the classification head provides no advantage; validating the multi-turn hierarchy requires multi-turn human data.

4.7 Perturbation Analysis

Under adversarial paraphrasing (BERTScore 0.92), classification-supervised variants achieve zero degradation on IID and OOD DD (Table 10) while non-adapted models degrade by $\sim$12 pp ($p < 0.001$); on ActorAttack, DAPT variants degrade ($\Delta = -20.4$ pp) but maintain 52 pp over vanilla (Appendix D).

4.8 Strategy Trajectory Analysis

The 9-class taxonomy provides interpretable characterization of known attacks (κ=0.78, n=50; Appendix E) but degrades to near-random on OOD (κ=0.126; Appendix X). Its value is twofold: as ablation infrastructure (scrambled labels produce equivalent detection, confirming label content is non-critical) and as an interpretability tool for characterizing known attacks; detection operates through domain-adapted `[CLS]` embeddings, not strategy accuracy.

5 Discussion

Topic shortcut and label equivalence. Three observations argue against a topic shortcut: unstable topic OOD (F1 = 0.541±0.207), scrambled labels within 3.9 pp, and label-free MLM detection (F1 $\geq$ 0.97).

Single-LLM confound. ToxicChat (+18–42 pp; Table 5) and cross-LLM experiments (Appendix H) confirm transfer. Mixed-LLM 9-class maintains DD F1 = 0.818±0.096 (+53 pp) while MLM collapses to vanilla level. A mixed-source training pilot preserves Qwen OOD detection with preliminary Llama transfer (Appendix I).

Variance and capacity. Classification supervision reduces variance on synthetic OOD (4–20×) but increases it on ToxicChat (std = 0.087 vs. 0.013); frozen probing localizes MLM instability to Stage-2 (Appendix S). Three encoders (125M–355M) differ in pretraining; same-architecture scaling is needed. TF-IDF collapses on ActorAttack (0.261 vs. 0.961 on DD).

Mechanism: discriminative compression. Representation analysis reveals the mechanism: classification DAPT compresses the feature space into few discriminative dimensions (participation ratio 3.7–9.4 vs. 25 for MLM and 32 for vanilla) while increasing class separability by 3–5 orders of magnitude (Fisher ratio 2.6–19.7 vs. <0.001; Appendix S). Weight distances from pretrained are similar across all variants ($\sim$0.2–0.4% relative), so the effect is representational restructuring, not parameter-space regularization. Scrambled labels produce comparable compression (Fisher = 2.6), confirming that classification structure—not label content—drives the effect (Chen et al., 2020; Grill et al., 2020; Zhang et al., 2021; Maennel et al., 2020). Frozen probing near-parity (AA gap 3.1 pp) and mean pooling matching BiGRU ($|\Delta| < 1$ pp) corroborate: discriminative structure is established in Stage-1, not Stage-2. This advantage is condition-specific: classification DAPT outperforms MLM only under multi-turn synthetic evaluation; on single-turn human data (ToxicChat), MLM generalizes better (Section 4.6).

Two-axis generalization: attack-side vs. benign-side. OOD F1 throughout this paper reflects attack-side generalization under IID benign conditions. The orthogonal challenge is benign-side shift: classification DAPT increases WildChat FPR from 48.1% (vanilla) to 68.1% (Appendix O), confirming adaptation amplifies benign sensitivity. Cross-LLM error decomposition attributes collapse to false positives on unfamiliar benign text. Ten target-domain conversations reduce FPR to 0% (95% CI upper 6.1%) without detection degradation (Appendix P).

6 Conclusion

Controlled ablation across thirteen encoder configurations decomposes multi-turn jailbreak generalization into two orthogonal axes: attack-side

(classification DAPT via discriminative compression of the feature space; unsupervised pretraining collapses under mixed-LLM data) and benign-side (10 target-domain conversations eliminate WildChat FPR 68.1% → 0%). Label sensitivity is attack-complexity-dependent (TOST equivalence on DD/FITD; $d = 0.44$ on ActorAttack); temporal order and label granularity are non-factors. Combined with 24.3× speedup over LLM-as-judge (Appendix V), this yields a deployment recipe requiring no target-domain attack data.

7 Limitations

All multi-turn findings derive from synthetic conversations generated by Qwen3-8B (350 training, 76–118 test per family); preliminary mixed-LLM training (Appendix I) suggests this limitation is addressable. ToxicChat validates transfer to human-authored single-turn attacks (+18–42 pp over vanilla), but no human-generated multi-turn jailbreak dataset exists. Whether the evidence hierarchy generalizes to human-authored multi-turn attacks remains the primary open question (length confound analysis in Appendix U).

OOD results reflect attack-side generalization only: benign data is IID, so OOD F1 figures are upper bounds (Section 5). On WildChat, 9-class FPR reaches 68.1% (Appendix O). This limitation is addressable: a benign adaptation curve (Appendix P) shows that as few as 10 target-domain benign conversations eliminate FPR without detection degradation, though validation on larger and more diverse OOD benign samples is needed. The 10-conversation benign adaptation result is validated only on WildChat (GPT-3.5 user conversations); whether it generalizes to other LLM assistants or enterprise chat systems is unevaluated. Residual calibration error (ECE = 0.126–0.141; Appendix R) suggests temperature scaling is needed for deployment.

Three encoders (125M–355M) differ in architecture and pretraining; same-architecture scaling is needed to isolate the capacity effect, and whether findings hold at LLM-scale encoders (7B+) remains open. Only persuasion-mediated multi-turn attacks fall within scope; single-turn token-level attacks (Zou et al., 2023), non-persuasive strategies, and non-text modalities are not addressed. Total pipeline cost is under 4 GPU-hours on a single A100.

Ethics Statement

This work is defensive: it improves detection of multi-turn jailbreak attacks rather than developing new attacks. We acknowledge dual-use risk, as the persuasion taxonomy could inform adversarial actors. We mitigate this by using only synthetic conversations (no real harmful content), withholding attack-generation prompts, and focusing contributions on detection mechanisms. No human subjects were involved; all data will be released under CC BY-NC 4.0.

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Table 6: LLM-as-judge jailbreak detection. IID: binary F1 (Crescendo test set); OOD: macro F1 (DD = Deceptive Delight, AA = ActorAttack). GPT-5.5 F1 values exclude unparseable responses and represent upper bounds.

Method	DD	AA	IID	Unparse.
Zero-shot				
GPT-5.5	–	–	.383	30–47%
Qwen3-8B	.244	.244	.100	0%
3-shot				
Qwen3-8B	.566	.652	–	0%
Trained classifier				
9-class (ours)	**.994**	**.972**	**1.000**	0%

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A LLM-as-Judge Baseline

Table 6 reports LLM-as-judge performance on IID and OOD data. On IID Crescendo, GPT-5.5 achieves precision of 1.0 at all turn prefixes but recall ranges from only 0.105 to 0.290, resulting in F1 = 0.383 on full conversations; 30–47% of responses are unparseable. Qwen3-8B performs worse (IID F1 = 0.100). On OOD data, zero-shot Qwen3-8B achieves macro F1 = 0.244 on both DD and ActorAttack by predicting nearly all conversations as benign. Even with 3-shot prompting, Qwen3-8B reaches only 0.566 (DD) and 0.652 (AA), while DisPeD maintains macro F1 $\geq$ 0.972. The poor performance of content-level reasoning on unseen attack families underscores the necessity of trained classifiers with domain-adapted representations.

B Full Cross-Attack Results by Prefix

C ActorAttack Full Results

Table 8 reveals three distinct prefix-length patterns. The 9-class model improves monotonically from K=1 to Full (+20 pp), consistent with cumulative evidence accumulation across turns. Vanilla DeBERTa *decreases* with more turns (0.337 at K=1 to 0.258 at Full), suggesting that additional attack turns push representations further into benign embedding regions. MLM peaks at K=3 (0.945) then declines to 0.893 at Full, indicating that unsupervised domain exposure captures early-turn patterns but is unstable under extended sequences.

C.1 Error Analysis

Per-sample majority vote across 3 seeds yields 98.3% accuracy on ActorAttack OOD (2 false negatives, 0 false positives). No sample is misclassified by all three seeds; all 7 seed-specific errors are false negatives concentrated in shorter conversations (3–6 turns). On these 7 error samples, vanilla achieves 0/3 correct votes per sample, while 9-class majority vote correctly classifies 5/7, confirming that domain adaptation provides robust per-sample coverage even where individual seeds fail.

D Perturbation Robustness

Table 9 shows perturbation behavior on IID data. Classification-supervised variants (9-class, binary, scrambled) achieve zero degradation at full conversation length regardless of label correctness. Models without jailbreak-domain exposure (Wikipedia MLM, vanilla) degrade by $\sim$12 pp. McNemar's test finds no significant pairwise differences among classification-supervised variants (all $p > 0.13$), though with $\leq$7 discordant pairs, statistical power is limited.

On OOD DD, the IID pattern replicates: classification-supervised variants show zero degradation while McNemar's test confirms 9-class vs. vanilla significance ($p < 0.001$, 49–78 discordant pairs). On OOD ActorAttack, domain-adapted variants degrade substantially (9-class $\Delta = -20.4 \pm 13.5$ pp, MLM -35.3 pp), but perturbation narrows rather than closes the adapted vs. non-adapted gap: 9-class perturbed F1 = 0.768 vs. vanilla 0.248 (McNemar $p < 0.001$). Domain specificity also manifests: Wikipedia MLM degrades by $\sim$12 pp while jailbreak-domain MLM degrades by only 2.6 pp.

E Strategy Trajectory Analysis

Figure 3 shows transition count matrices for three attack families. Crescendo conversations exhibit broad transition coverage across all nine strategy categories (1,818 total transitions), with the highest concentration in Role Assignment self-transitions (545); this reflects its gradual escalation design. FITD conversations show a markedly sparser pattern (541 transitions), with Role$\rightarrow$Obfuscation (161) as the dominant transition, consistent with FITD's commitment-consistency exploitation (Cialdini, 2001). DD conversations span 5 active source strategies (565 transitions), with Emotion$\rightarrow$Obfuscation (129) as the dominant tran-

Table 7: Full cross-attack generalization results (F1) at all prefix lengths. 3-seed mean±std where available. "–" indicates insufficient conversation length for that prefix (FITD conversations contain 2–3 turns).

Method	K=1	K=2	K=3	K=5	Full
IID (Crescendo)					
9-class (ours)	1.000±.000	1.000±.000	1.000±.000	1.000±.000	1.000±.000
MLM-only	.960	.974	.987	1.000	1.000
Vanilla DeBERTa	.968±.033	.978±.016	.987±.000	.982±.014	1.000±.000
TF-IDF LR	1.000	1.000	1.000	1.000	1.000
OOD (Deceptive Delight)					
9-class (ours)	.990±.013	.981±.013	.972±.026	.997±.005	.994±.005
Binary	.994±.009	.994±.009	.984±.008	1.000±.000	1.000±.000
Scrambled	.997±.005	.981±.026	.975±.029	.997±.005	.994±.005
MLM-only	.973±.019	.993±.009	.997±.005	.997±.005	.914±.115
Vanilla DeBERTa	.628±.038	.479±.038	.389±.027	.351±.030	.312±.026
TF-IDF LR	1.000	1.000	.994	1.000	.961
OOD (FITD)					
9-class (ours)	1.000	1.000	–	–	1.000
Vanilla DeBERTa	.990	1.000	–	–	1.000
TF-IDF LR	1.000	–	–	–	1.000

Table 8: Full ActorAttack OOD results (F1) at all prefix lengths. 3-seed mean±std (Binary per-K reports n=2; Full is 3-seed).

Method	K=1	K=2	K=3	K=5	Full
DeBERTa-v3					
Binary	.788±.020	**.935±.018**	**.976±.005**	**.995±.005**	**.990±.008**
9-class	.773±.041	.855±.048	.921±.046	.966±.028	.972±.026
Scrambled	.652±.034	.804±.053	.854±.046	.910±.052	.951±.043
MLM-only	.736±.150	.909±.058	.945±.039	.929±.033	.893±.080
Wiki MLM	.656±.224	.672±.252	.693±.280	.689±.285	.686±.297
Topic	.513±.077	.580±.109	.565±.100	.615±.111	.625±.141
Vanilla	.337±.049	.306±.026	.284±.011	.262±.006	.258±.000
TF-IDF LR	.633	.730	.571	.242	.261
RoBERTa					
9-class	**.738±.033**	.805±.033	.884±.066	.925±.066	.939±.046
MLM	.633±.133	.665±.150	.713±.174	.719±.184	.730±.201
Vanilla	.410±.126	.425±.143	.432±.144	.466±.144	.466±.163

sition, aligning with DD's design of embedding harmful content within emotionally engaging contexts (Palo Alto Networks Unit 42, 2024).

F Centered CKA Heatmaps

G Embedding Space Visualization

Figure 5 visualizes mean-pooled `[CLS]` embeddings via t-SNE for three representative variants (Vanilla, JB-MLM, 9-Class) on DD OOD and ActorAttack. Vanilla embeddings show dispersed jailbreak clusters with ambiguous decision boundaries, while domain-adapted variants (JB-MLM, 9-Class) produce tighter, more separable clusters. JB-MLM achieves the clearest bimodal separation on both evaluation sets, consistent with its superior detection performance.

H Cross-LLM Validation

Table 11 shows three inference conditions. When both jailbreak and benign text come from Llama (row 2), all variants collapse (F1 = 0.26–0.43), but this conflates two sources of distribution shift: unfamiliar jailbreak patterns and unfamiliar benign text. Row 3 isolates jailbreak transferability by pairing Llama jailbreaks with the original Qwen benign distribution. Under this condition, 9-class recovers to F1 = 1.000 on DD and 0.801 on ActorAttack, and JB-MLM reaches 0.903 on ActorAttack. The contrast between rows 2 and 3 shows that the dominant failure mode in row 2 is benign distribution mismatch (false positives from unfamiliar benign text), not inability to recognize cross-LLM jailbreak patterns. Vanilla remains at 0.26–0.31 across all conditions, further confirming that en-

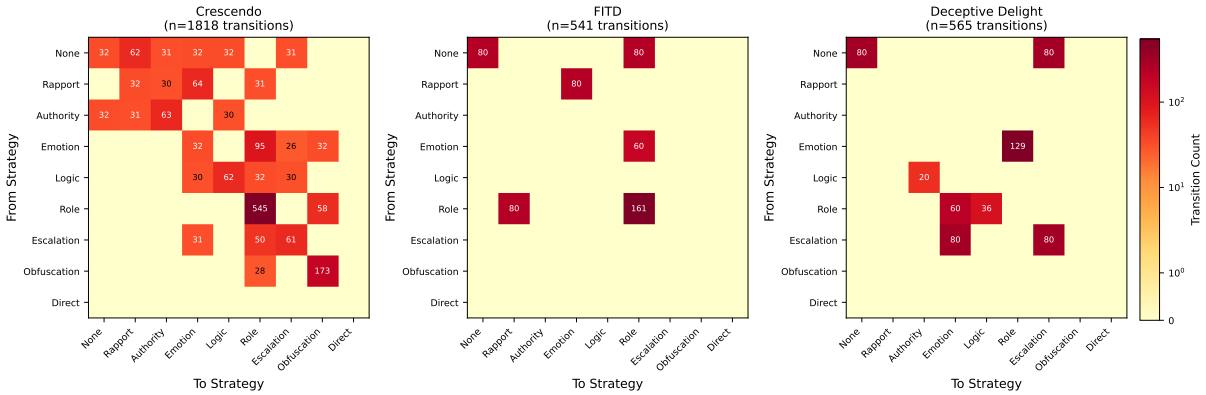


Figure 3: Persuasion strategy transition count matrices for three attack families, based on ground-truth intended strategy labels. Strategy labels validated on n=50 samples with a single annotator (κ=0.78); fine-grained trajectory patterns should be interpreted with this limitation. Crescendo exhibits broad coverage across all strategy categories, while FITD and DD concentrate transitions in smaller subsets. Row i, column j counts transitions from strategy i at turn t to strategy j at turn $t+1$.

Table 9: Perturbation behavior under adversarial paraphrasing on IID Crescendo (n=76). Δ: F1 degradation (pp). Classification-supervised variants achieve zero degradation at full length; this pattern does not generalize to ActorAttack.

Method	Δ Full	Δ K=1	Δ K=2	Δ K=3
9-class (ours)	**0.0**	**−2.6**	**0.0**	**0.0**
Scrambled	**0.0**	−5.3	−2.6	**0.0**
Binary	**0.0**	−9.2	−2.6	**0.0**
TF-IDF LR	−1.3	−9.6	−4.1	−1.3
MLM-only	−2.6	−9.2	−4.0	−4.0
Topic (seed 42)	−4.0	−2.7	−2.7	**0.0**
Wikipedia MLM	−12.1	−12.4	−15.4	−17.9
Vanilla DeBERTa	−11.8	−14.6	−12.1	−13.8

Table 10: Perturbation robustness on OOD Deceptive Delight (n=118). Δ: F1 change (pp) from clean to paraphrased. 3-seed mean±std where available.

Variant	Clean Full	Pert. Full	Δ Full	Δ K=1
9-class	.994±.005	1.000±.000	+0.6	+0.6
Binary	1.000±.000	1.000±.000	0.0	+0.6
Scrambled	.994±.005	1.000±.000	+0.6	−0.3
JB MLM	.914±.115	.941	+2.7	0.0
Wiki MLM*	.904±.115	.872	−3.3	−3.2
Vanilla†	.312±.026	.339±.055	+2.7	−6.8±1.1

*Wiki MLM degradation is seed-driven: seed 42 accounts for −9.8 pp; other seeds show 0.0 pp change.
†Vanilla Δ Full = +2.7 pp reflects floor-effect noise (baseline 0.312, std 0.055), not meaningful improvement.

coders without DAPT lack OOD detection capability regardless of the LLM source. The residual degradation on ActorAttack (0.801 vs. 0.972) reflects genuine cross-LLM variation in structurally complex attack patterns.

Table 12 decomposes DD Full F1 into true positive and false positive rates. TP rates are identical regardless of the benign source: domain-adapted models detect Llama-generated jailbreaks at the same rate whether evaluated against Qwen or Llama benign text. The F1 collapse in the full Llama condition (Table 11, row 2) is driven entirely by false positives on unfamiliar benign text (FP rate: 88–90% on Llama vs. 0–0.9% on Qwen for adapted variants). The model implicitly treats non-Qwen text style as a jailbreak indicator, a spurious correlation from single-LLM training data. This failure reflects a data diversity limitation, not a methodological one; mixed-LLM training (Table 13) partially addresses this confound.

I Mixed-LLM Training Results

Table 13 reports detection performance when training on a mixed corpus of Qwen3-8B and Llama-3.1-8B conversations. Classification-based DAPT (9-class) maintains the strongest DD OOD performance (0.818±0.096), outperforming both vanilla (0.291±0.065) and MLM (0.289±0.101) by 53 pp. ActorAttack remains challenging under mixed-LLM conditions (9-class F1 = 0.504±0.107), suggesting structurally complex attacks require larger or more diverse training data. ToxicChat performance shows no significant improvement over single-source training (0.550±0.052 vs. 0.596±0.087; Section 4.6), consistent with ToxicChat being zero-shot single-turn transfer rather than a standard OOD benchmark.

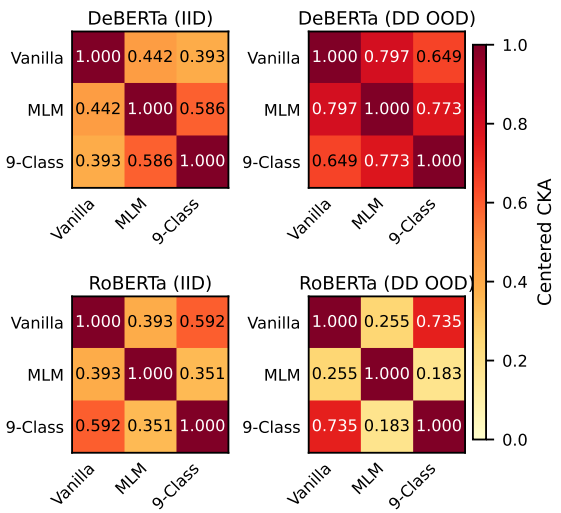


Figure 4: Centered CKA similarity (Kornblith et al., 2019) between encoder variants on IID and OOD data for DeBERTa-v3 and RoBERTa. Low off-diagonal values (vanilla vs. adapted variants) confirm that continued pretraining substantially reshapes representations; the MLM vs. 9-class block shows distinct geometries (CKA ≤ 0.2 on RoBERTa OOD).

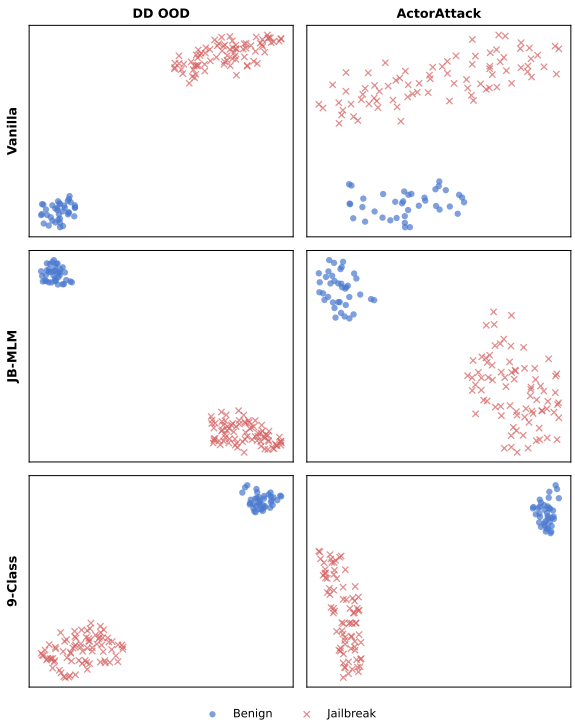


Figure 5: t-SNE visualization of mean-pooled `[CLS]` embeddings (perplexity=30) for Vanilla, JB-MLM, and 9-Class variants on DD OOD and ActorAttack (118 samples each; blue=benign, red=jailbreak).

Table 11: Cross-LLM validation (F1, full conversation). All models are trained on Qwen3-8B data; conditions vary the source of jailbreak (JB) and benign (ben.) conversations at inference. Values are evaluated under a matched 50-sample subset protocol and differ from Table 3, which uses the full evaluation set.

	DD Full			*AA Full*		
Condition	Van.	MLM	9-cl.	Van.	MLM	9-cl.
Qwen JB + Qwen ben.	.312	.914	.997	.258	.893	.972
Llama JB + Llama ben.	.261	.282	.433	.261	.368	.321
Llama JB + Qwen ben.	.302	.778*	1.000	.302	.903	.801

*High seed variance (std = 0.306): seed 456 outlier (F1 = 0.345); remaining two seeds achieve F1 $\geq$ 0.97.

Table 12: Error decomposition for DD Full cross-LLM conditions (matched 50-sample subsets). TP rate is identical across benign conditions; the F1 collapse in the Llama-only condition (Table 11, row 2) is driven entirely by false positives on unfamiliar benign text.

Metric	Van.	MLM	9-cl.
TP rate (Llama JB)	0.0%	68.0%	100.0%
FP rate (Llama ben.)	29.3%	88.0%	90.0%
FP rate (Qwen ben.)	0.0%	0.9%	0.0%

Cross-LLM training pilot. To test whether mixed-source training preserves Qwen OOD detection while enabling cross-LLM transfer, we train 9-class DAPT on a combined corpus: 175 Qwen jailbreak + 175 Qwen benign + 164 Llama-3.1-8B-Instruct jailbreak + 150 WildChat benign ($n = 664$ total; 3-seed mean). Qwen OOD detection is fully preserved (DD/AA/FITD F1 = 1.000 across all families), and WildChat FPR remains 0%. On held-out Llama-generated conversations, Crescendo recall reaches 100% ($n = 40$; 95% CI lower bound 91.2%). Llama DD, AA, and FITD each achieve 100% recall, but with only $n = 10$ per family the 95% CI lower bound is 72.2%, limiting the strength of this evidence. These results are preliminary: they confirm that mixed-LLM training does not degrade existing detection and provide initial evidence of cross-LLM transfer, but definitive conclusions require larger Llama-generated test sets.

J Dataset Scale Validation

Table 14 validates robustness to dataset scale by doubling the training corpus from 500 to 1,000 conversations with new topic seeds. The hierarchy 9-class $\gg$ MLM $\approx$ vanilla holds at 2$\times$ scale: classification-based DAPT maintains 40–56 pp advantage over vanilla, while MLM provides no improvement over vanilla on either attack family (DD: 0.338 vs. 0.326; AA: 0.351 vs. 0.508).

Table 13: Mixed-LLM training (250 Qwen + 250 Llama conversations, 3-seed mean±std where available). ToxicChat evaluated on balanced set (408 conversations).

Variant	Mixed	DD OOD	AA OOD	FITD OOD	ToxicChat
9-class	**.996±.006**	**.818±.096**	**.504±.107**	**.997±.005**	**.550±.052**
MLM	.681±.226	.289±.101	.258±.075	.679±.188	.256±.053
Vanilla	.802±.010	.291±.065	.389±.073	.867±.022	.356±.009

Table 14: Dataset scale validation (1,000 Qwen conversations, 3-seed mean±std). IID results omitted due to train/test overlap (71% of original test set present in expanded training data).

Variant	DD OOD	AA OOD
9-class	**.889±.041**	**.913±.041**
MLM	.338±.010	.351±.018
Vanilla	.326±.000	.508±.085

K RoBERTa-large DAPT Learning Rate Control

Table 15: RoBERTa-large (355M) DAPT learning rate control on OOD Full (3-seed mean±std). Reducing the learning rate worsens both DD and AA, ruling out over-adaptation as the cause of the regression in Table 4.

Config	DD Full	AA Full
Vanilla (no DAPT)	.927±.036	1.000±.000
9-class, LR=2×10^{-5}	.987±.004	.946±.057
9-class, LR=1×10^{-5}	.948±.034	.781±.054

Reducing the DAPT learning rate from 2×10^{-5} to 1×10^{-5} worsens both DD and AA performance, which rules out over-adaptation as the cause of the regression observed in Table 4. An MLM-only variant (without classification) produces the same pattern (AA Full = 0.874±0.014); representation drift from continued pretraining, not classification-specific bias, is the underlying cause. Domain-adaptive fine-tuning narrows the representation space of a sufficiently large encoder and reduces its inherent OOD generalization ability.

L TOST Equivalence Sensitivity Analysis

On DD, TOST confirms scrambled ≈ correct labels even at δ=2 pp (p=0.035 at n=3), demonstrating that label correctness is not a detection factor; the DD equivalence conclusion is robust to both margin choice and seed count ($p < 10^{-8}$ at n=10). On FITD, equivalence is similarly confirmed ($p < 10^{-8}$, $d \approx 0$ at n=10). On ActorAttack, extending from n=5 to n=10 reveals a moderate effect: Cohen's d increases from 0.09 to 0.44 (medium), with 9-class outperforming scrambled by 3 pp (0.979 vs. 0.949). TOST remains inconclusive (p = 0.183 at δ=5 pp), and variance is driven by seed 314 (scrambled F1 = 0.833). Label correctness is thus attack-complexity-dependent: negligible on template-based families (DD, FITD) but moderate on structurally complex attacks (ActorAttack).

Table 16: TOST equivalence test sensitivity to margin δ (scrambled vs. 9-class). Rows grouped by seed count: n=3 (42, 123, 456), n=5 (+789, 1000), n=10 (+2024, 2025, 2026, 314, 628). p-values from two one-sided t-tests; $p < 0.05$ confirms equivalence within $\pm\delta$.

Dataset	n	δ=5 pp	δ=3 pp	δ=2 pp
DD OOD	3	.006 ✓	.016 ✓	.035 ✓
DD OOD	5	<.001 ✓	<.001 ✓	.002 ✓
DD OOD	10	$<10^{-8}$ ✓	$<10^{-8}$ ✓	$<10^{-8}$ ✓
FITD OOD	10	$<10^{-8}$ ✓	$<10^{-8}$ ✓	$<10^{-8}$ ✓
ActorAttack	3	.290	.428	.507
ActorAttack	5	.080	.200	.310
ActorAttack	10	.183	—	—

M Reproducibility Details

Stage-1 training. All variants use DeBERTa-v3-base (184M parameters) with learning rate 2×10^{-5}, AdamW optimizer (β_1=0.9, β_2=0.999, weight decay 0.01), linear warmup over the first 10% of steps, batch size 16, and 4 epochs. MLM variants use 15% random token masking. Training takes approximately 30 minutes per variant on a single NVIDIA A100 (40GB).

Stage-2 training. The BiGRU has hidden dimension 256, 2 layers, and dropout 0.3 (bidirectional, so output dimension 512). The MLP classifier has layers $512 \rightarrow 256 \rightarrow 2$ with ReLU activation. Training uses learning rate 1×10^{-3}, Adam optimizer, batch size 32, and 20 epochs with early stopping (patience 5) on validation F1. Stage-2 training completes in under 5 minutes per variant.

Data generation. Jailbreak conversations are generated using Qwen3-8B with attack-specific prompt templates. Crescendo conversations follow a 5–8 turn gradual escalation structure. FITD conversations use a 2–3 turn commitment-consistency structure. Deceptive Delight conversations embed harmful content within emotionally engaging 3–5 turn sequences. Benign conversations are generated with topic-anchored prompts across five domains (technology, health, finance, education, gen-

eral) with matched turn counts.

Strategy annotation. Each user turn is annotated by Qwen3-8B using a structured prompt that provides definitions and two examples for each of the nine strategy categories. The prompt instructs the model to output a single category label. We plan to release all generation and annotation prompts, the full dataset, and training code upon acceptance.

Adversarial paraphrasing. The paraphrased test set is generated by instructing Qwen3-8B to rephrase each user turn while preserving the communicative intent. The resulting paraphrases have mean BERTScore 0.92 and mean lexical overlap 0.20 with the originals. These metrics confirm substantial surface-level variation.

Cross-checkpoint variance. To disentangle Stage-1 from Stage-2 variance, we evaluate three independently trained 9-class DAPT checkpoints (seeds 42, 123, 456) with a fixed downstream BiGRU seed (42). Results: IID F1 = 1.000±0.000, DD OOD = 0.926±0.026, AA OOD = 0.935±0.013. Stage-1 training is stable (OOD std $\leq$ 2.6 pp), confirming that the variance reported in the main paper is not dominated by Stage-1 initialization.

N End-to-End Fine-Tuning Upper Bound

Table 17: Frozen probing vs. end-to-end (E2E) fine-tuning for DeBERTa-v3-base 9-class (3-seed mean±std). E2E unfreezes the Stage-1 encoder during Stage-2 training. Frozen values use main-experiment checkpoints (matching Tables 2–3); Table 23 uses plan_002 checkpoints, yielding higher frozen accuracy (DD = .997, AA = .972) and a narrower E2E gap ($\leq$2.5 pp).

Config	IID	DD OOD	AA OOD
Frozen probing	1.000±.000	.994±.005	.939±.047
E2E fine-tuning	1.000±.000	1.000±.000	**.997±.005**
Δ	.000	+.006	+.058

End-to-end fine-tuning unfreezes the Stage-1 encoder during Stage-2 BiGRU training, allowing gradient flow through the full pipeline. On IID and OOD DD, the two configurations are indistinguishable; on structurally complex ActorAttack, E2E recovers +5.8 pp. The modest gain confirms that frozen probing captures the vast majority of discriminative information encoded by Stage-1 DAPT, and that the remaining OOD gap in the frozen setting reflects downstream optimization constraints rather than missing representational capacity.

O Real-World False Positive Rate

Table 18: False positive rate (%) on 226 WildChat benign multi-turn conversations (GPT-3.5 users, avg 7.4 turns per conversation). All models trained on synthetic Qwen3-8B data (1,000-conversation scale). 3-seed mean±std.

Variant	Seed 42	Seed 123	Seed 456	Mean±Std
9-class	66.8	61.1	76.5	68.1±6.4
MLM	36.7	64.6	32.3	44.5±14.3
Vanilla	47.3	42.5	54.4	48.1±4.9

Table 18 evaluates false positive rates on 226 WildChat benign multi-turn conversations from real GPT-3.5 users (avg 7.4 turns). All variants produce FPR far exceeding deployable thresholds: 9-class is worst at 68.1%, followed by vanilla (48.1%) and MLM (44.5%). Domain adaptation *increases* FPR relative to vanilla (+20 pp for 9-class), indicating that continued pretraining amplifies dependence on the training benign distribution. This is consistent with the cross-LLM false positive spike (Table 12): models trained on synthetic Qwen3-8B benign text implicitly treat unfamiliar conversational styles as jailbreak indicators. High MLM seed variance (std = 14.3%) suggests that unsupervised pretraining is particularly unstable under benign distribution shift. Deployment requires per-domain benign calibration or mixed-source benign training data.

Post-hoc calibration. ROC AUC = 0.700±0.054 indicates insufficient discrimination between WildChat benign and jailbreak conversations at the representation level. At TPR$\geq$95%, the optimal threshold still yields FPR = 62.7%—far above deployable levels. Temperature scaling cannot improve this trade-off, as ROC curves are invariant to monotonic score transforms. The AUC ceiling reflects genuine domain shift: WildChat benign and jailbreak representations overlap substantially, and no post-hoc calibration can overcome this; in-domain benign adaptation is required.

P Benign Distribution Diversification Pilot

To test whether the high WildChat FPR (Section O) reflects a fundamental limitation or a remediable

distribution mismatch, we conduct a pilot retraining experiment mixing 150 WildChat benign conversations into the training set. We filter 27 of 226 WildChat conversations containing jailbreak-indicative patterns, use 150 for training and 49 for held-out evaluation. Both training and evaluation WildChat samples are drawn from the same source (GPT-3.5 users); this is therefore an IID evaluation of benign coverage, not an OOD generalization test.

Class ratio shift. Adding 150 WildChat benign to the original 350-conversation training set (175 jailbreak + 175 synthetic benign) shifts the benign-to-jailbreak ratio from 50:50 to approximately 65:35 (325 benign vs. 175 jailbreak). To control for this confound, we downsample WildChat benign to 88 conversations (175 jailbreak : 88 WildChat benign + 87 original benign $\approx$ 50:50, $n = 350$). This balanced variant achieves identical WildChat FPR (0%, same CI) while maintaining OOD detection (AA F1 = 0.991, FITD/DD F1 = 1.000), confirming that class ratio is not the driver of FPR reduction.

Table 19: Effect of benign distribution diversification on WildChat FPR and OOD jailbreak detection. Mixed training adds 150 WildChat benign to the original 350 synthetic conversations. 3-seed mean (seeds 42, 123, 456). WildChat evaluation is IID (same source as training WildChat).

Metric	**Original**	**Mixed**
WildChat FPR (%)	68.1	**0.0**
IID F1	1.000	1.000
DD OOD F1	1.000	1.000
AA OOD F1	∼0.960	0.956
FITD OOD F1	1.000	1.000

Mixed benign training reduces WildChat FPR from 68.1% to 0% (exact binomial 95% CI upper bound: 6.1%, $n = 49$) with no degradation in cross-attack detection (ActorAttack F1 = 0.956, within 0.4 pp of original). This confirms the diagnosis from Section O: when representative benign data is available, the high FPR is remediable without sacrificing detection performance. The evaluation set is small ($n = 49$) and IID with respect to the training WildChat distribution, so the result should be interpreted as evidence that benign distribution mismatch—not model capacity—is the bottleneck; validation on larger and independently sourced OOD benign corpora is needed to establish deployment-ready FPR bounds.

Benign adaptation curve. To quantify how many OOD benign samples suffice, we vary the number of WildChat benign conversations added to training ($N \in \{0, 5, 10, 25, 50, 100, 150\}$) and retrain the full pipeline (9-class DAPT + BiGRU) for each N (seed 42).

Table 20: Benign adaptation curve: WildChat FPR and OOD detection F1 as a function of N WildChat benign conversations added to the 350-conversation training set. Seed 42. WildChat evaluation $n = 49$.

N	Train	FPR (%)	DD F1	AA F1	FITD F1
0	350	65.3	1.000	0.981	1.000
5	355	16.3	0.990	0.981	1.000
10	360	**0.0**	1.000	1.000	1.000
25	375	0.0	1.000	0.953	1.000
50	400	0.0	1.000	1.000	1.000
100	450	0.0	1.000	1.000	1.000
150	500	0.0	1.000	0.981	1.000

FPR drops sharply from 65.3% ($N = 0$) to 16.3% ($N = 5$) to 0% ($N = 10$), remaining at 0% for all $N \geq 10$. OOD detection is preserved throughout: all F1 scores ≥ 0.953 (minor ActorAttack variation at $N = 25$ is within seed noise). This demonstrates that benign distribution mismatch is addressable with minimal annotation effort—10 conversations from the target domain suffice to eliminate false positives entirely (95% CI upper bound: 6.1% at $n = 49$). IID F1 remains 1.000 for all N. The curve is single-seed; multi-seed validation is left to future work.

Q BERT-base Proxy Baseline

Table 21: BERT-base-uncased (110M) frozen probing without DAPT, matching DeepContext's encoder family (Albrethsen et al., 2026). 3-seed mean±std. DeBERTa-v3 results from Table 4 for comparison.

Encoder	IID	DD OOD	AA OOD
BERT-base (110M), no DAPT	.996±.006	.318±.031	.275±.027
DeBERTa-v3 (184M), no DAPT	1.000±.000	.312±.043	.258±.000
DeBERTa-v3 (184M), 9-class	1.000±.000	.994±.005	.939±.047

Table 21 compares BERT-base-uncased (110M parameters, matching DeepContext's encoder) against DeBERTa-v3-base (184M), both without domain-adaptive pretraining. The two vanilla encoders achieve comparable OOD performance (DD: 0.318 vs. 0.312; AA: 0.275 vs. 0.258), despite a 1.7× parameter difference and different pretraining procedures. Both collapse on OOD evaluation while 9-class DAPT recovers to 0.994 (DD) and 0.939 (AA). This confirms that cross-attack

generalization is driven by domain adaptation, not encoder architecture or capacity at this scale.

R Confidence Calibration

Table 22: Expected calibration error (ECE, 10 bins) for 9-class and vanilla on IID and OOD evaluation (3-seed mean±std). Lower is better.

Variant	IID	DD OOD	AA OOD
9-class	≈0.000	0.143±0.053	0.116±0.053
Vanilla	0.009±0.007	0.669±0.004	0.504±0.066

The 9-class model achieves substantially lower ECE than vanilla on OOD data (0.12–0.14 vs. 0.50–0.67), indicating that domain adaptation improves not only discriminative performance but also confidence calibration. IID calibration is near-perfect for both variants. The residual OOD ECE (0.12–0.14) suggests that temperature scaling could further improve deployment reliability.

S Frozen Probing Verification

Table 23: Frozen probing vs. end-to-end (E2E) fine-tuning (3-seed mean±std). 9-class: plan_002 checkpoints; MLM: plan_017_mlm_control (matched training). Frozen probing trains only BiGRU+MLP on frozen DAPT embeddings.

Config	IID	DD OOD	AA OOD	FITD OOD
Frozen 9-class	1.000±.000	.997±.005	.972±.026	1.000±.000
Frozen MLM	.982±.016	.993±.009	.941±.037	.987±.009
E2E 9-class	1.000±.000	1.000±.000	.997±.005	—

Table 23 compares frozen probing across DAPT objectives and against end-to-end fine-tuning. On easy OOD (DD, FITD), frozen MLM and frozen 9-class representations perform comparably ($\Delta \leq 1.3$ pp). On hard OOD (ActorAttack), 9-class retains a modest advantage (0.972 vs. 0.941, $\Delta = 3.1$ pp) with lower variance ($\sigma = 0.026$ vs. 0.037). Both far exceed vanilla encoders ($\leq$0.318; Table 21). The E2E advantage over frozen 9-class is $\leq$2.5 pp, confirming that Stage-1 representation quality is the primary source of OOD generalization.

A striking asymmetry emerges when comparing frozen probing to the full pipeline (Table 3): MLM frozen representations *outperform* the full MLM pipeline on AA OOD (0.941 vs. 0.893), while 9-class frozen exactly matches its full pipeline (0.972 vs. 0.972). This indicates that MLM's cross-seed instability originates in Stage-2 BiGRU training rather than Stage-1 DAPT—the MLM encoder produces adequate representations that are then degraded by unstable downstream optimization. Classification DAPT, by contrast, produces representations that remain robust through Stage-2 training, explaining its consistently lower variance (Table 3).

This two-level structure—DAPT vs. no-DAPT producing a representation quality leap (+70 pp), classification vs. MLM DAPT producing a stability difference—is consistent with two further observations: λ=0 matches $\lambda > 0$ during E2E fine-tuning (Table 30), and mean pooling matches BiGRU under E2E (Table 25). Once DAPT representations are strong, downstream configuration details have minimal impact.

Representation geometry analysis. To characterize the mechanism behind classification DAPT's robustness, we analyze Stage-1 representations across all seven DeBERTa-v3-base variants (seed 42). We extract `[CLS]` embeddings from 150 test conversations (50 IID + 50 DD OOD + 50 AA OOD) and compute three metrics.

Table 24: Representation geometry and class separability across Stage-1 variants. PR: participation ratio (effective dimensionality of the embedding space). Fisher ratio: inter-class distance^2 / sum of intra-class variances. Weight Δ: L2 distance from pretrained weights (relative to pretrained norm).

Variant	PR	Fisher	Wt. Δ (%)
Vanilla	31.9	<0.001	0.000
Wiki-MLM	25.0	<0.001	0.221
JB-MLM	24.6	<0.001	0.256
Topic	14.5	0.012	0.371
Scrambled	7.8	2.60	0.283
9-class	9.4	4.47	0.303
Binary	3.7	19.7	0.239

Classification variants compress representations into 2–9 effective dimensions (vs. 25–32 for MLM/vanilla) while increasing Fisher ratio by 3–5 orders of magnitude. Weight distances from pretrained are comparable across all variants (∼0.2–0.4%), indicating that the mechanism is representational restructuring rather than parameter-space regularization. Scrambled labels produce similar compression (PR = 7.8, Fisher = 2.6), confirming that the classification objective structure—not label semantics—drives discriminative compression.

Table 25: Temporal aggregation comparison: mean pooling vs. BiGRU under frozen and end-to-end (E2E) settings (3-seed mean±std). Plan_002 9-class checkpoints.

Setting	Aggregator	DD OOD	AA OOD
Frozen	BiGRU	.997±.005	.972±.026
	Mean pooling	.991±.013	.962±.015
E2E	BiGRU	1.000±.000	.997±.005
	Mean pooling	1.000±.000	1.000±.000

T Temporal Aggregation Comparison

Table 25 compares BiGRU and mean pooling under both frozen and E2E settings. Under frozen probing (the main paper's pipeline), BiGRU provides a marginal advantage on ActorAttack (+1.0 pp) with overlapping confidence intervals, consistent with the main text finding that temporal order is not a core detection factor ($|\Delta| < 1$ pp under turn shuffling; Section 4.3). Under E2E fine-tuning, the difference vanishes entirely: mean pooling saturates all benchmarks at F1 = 1.000. The choice of frozen BiGRU in the main pipeline reflects a design trade-off: it enables representation-level analysis (CKA, t-SNE) that E2E would confound, while sacrificing $\leq$1 pp on the hardest OOD set.

U Length Stratification Analysis

Table 26: Jailbreak recall by conversation length bin on OOD evaluation (3-seed majority vote). Benign conversations span 12–20 turns; jailbreak conversations span 3–10 turns.

Dataset	Variant	1–3 turns	4–6 turns	7–10 turns
DD OOD	9-class	—	100.0%	99.4%
	Vanilla	—	$\leq$2.6%	$\leq$1.7%
AA OOD	9-class	93.3%	95.6%	100.0%
	Vanilla	0.0%	$\leq$2.6%	0.0%

Table 26 stratifies jailbreak recall by conversation length. The 9-class model shows no systematic degradation: ActorAttack recall increases monotonically from 93.3% (1–3 turns) to 100% (7–10 turns), and DD recall is near-perfect across all bins.

Length confound. Benign conversations (12–20 turns) and jailbreak conversations (3–10 turns) have zero length overlap in the OOD evaluation sets, raising the concern that the model discriminates by conversation length rather than content. Three observations argue against this interpretation: (1) Vanilla DeBERTa has access to identical length information via the BiGRU but achieves recall $\leq$2.6% across all bins—length alone is insufficient without domain-adapted representations. (2) At prefix K=1, where all conversations are length-equalized, 9-class still achieves F1 = 0.773±0.041 on ActorAttack (Table 8), demonstrating content-based detection independent of length. (3) The model is trained on IID data (Crescendo) and cannot learn OOD-specific length distributions; generalization to unseen length patterns must rely on content features that transfer across attack families.

Table 27: Turn-count independence verification (9-class, plan_002 checkpoints, 3 seeds). Part A: FPR on test benign truncated to K user turns (all seeds FPR = 0). Part B: WildChat FPR by turn-count bucket (mean±std). Part C: jailbreak detection (F1) after padding with benign turns to 15+ total turns (mean±std where available).

Condition	FPR / F1	n
Part A: Benign truncation (test set)		
Original (12–20 turns)	FPR = .000	38
Truncated K=3	FPR = .000	38
Truncated K=5	FPR = .000	38
Truncated K=8	FPR = .000	38
Truncated K=10*	FPR = .000	38
Part B: WildChat FPR by turn count		
Short (3–10 turns)	FPR = .770±.105	196
Medium (11–15 turns)	FPR = .942±.054	23
Long (16+ turns)	FPR = .857±.117	7
Part C: Jailbreak padding to 15+ turns		
DD original	F1 = .996	80
DD padded	F1 = .971 (Δ=−2.5 pp)	80
AA original	F1 = .963	80
AA padded	F1 = .383±.261 (Δ=−57.9 pp)	80

* K=10 did not truncate any conversation (test benign max ∼8 user turns).

Direct empirical verification. Truncating benign conversations to match jailbreak turn counts (Part A) produces zero false positives across all truncation levels and all 3 seeds, directly confirming that the model does not use conversation length as a detection shortcut. K=3 and K=5 are the effective control conditions where all 38 conversations were truncated; K=10 exceeds the maximum test benign length and serves as a no-op control.

WildChat FPR by turn count (Part B) provides complementary evidence: short conversations (3–10 turns, n=196) have *lower* FPR than medium-length ones (11–15 turns, n=23), inconsistent with a "short = jailbreak" shortcut. Medium and long buckets have limited sample sizes and should be

interpreted as directional evidence only.

Padding jailbreak conversations with benign turns (Part C) reveals an asymmetry: DD detection is resilient (Δ=$-$2.5 pp), but ActorAttack collapses (Δ=$-$57.9 pp). Padding experiments append benign turns *after* the complete attack sequence; in practice, adversaries would not dilute their own successful attacks. The collapse nonetheless exposes BiGRU temporal aggregation as vulnerable to attention dilution on subtle attack signals, a limitation discussed in Section 7.

V Efficiency

Table 28: Inference efficiency comparison on 100 conversations (50 IID + 50 DD OOD), same GPU (RTX PRO 6000). All methods measured under identical conditions.[†]

Method	Latency (ms)	Params	VRAM
DeBERTa+BiGRU (ours)	13.1±1.6	186.7M	868 MB
Qwen3-8B 3-shot	318.5±26.8	8.2B	16.1 GB

[†]LLM inference speed varies significantly across GPU generations; same-hardware comparison ensures a fair speedup estimate.

Table 28 compares inference cost under identical conditions. DeBERTa+BiGRU processes a conversation in 13.1 ms, 24.3× faster than Qwen3-8B 3-shot (318.5 ms) while achieving substantially higher OOD detection (DD = 0.566, AA = 0.652 for Qwen3-8B; Table 6). The encoder pipeline requires 868 MB GPU memory (18.5× less than the LLM), enabling deployment on consumer hardware.

V.1 Data Efficiency

Table 29: Data efficiency of 9-class DAPT (3-seed mean±std, sample std with $n-1$ divisor) and vanilla baseline (seed 42 only). N = total training conversations (balanced jailbreak/benign).

N	**Variant**	IID	DD OOD	AA OOD
50	9-class	1.000±.000	1.000±.000	.969±.037
125	9-class	1.000±.000	1.000±.000	.975±.019
250	9-class	1.000±.000	1.000±.000	.975±.035
350	9-class	1.000±.000	.978±.030	.972±.033
50	Vanilla	.646	.529	.234
125	Vanilla	.532	.244	.244
250	Vanilla	.987	.365	.502
350	Vanilla	1.000	.360	.489

Table 29 shows that 9-class DAPT reaches near-saturation performance with as few as 50 conversations (25 jailbreak + 25 benign): AA OOD F1 = 0.969±0.037, already surpassing vanilla at full training size (N=350, AA = 0.489) by 48 pp using only 14% of the data. DD OOD is perfectly solved at all training sizes. Vanilla performance is noisy across N (single seed); the consistent pattern is that vanilla never exceeds DD = 0.53 or AA = 0.50 regardless of training size, while 9-class maintains AA $\geq$ 0.969 at all N. Seed 456 is systematically lower across all N values, contributing most of the cross-seed variance; this is consistent with the initialization sensitivity observed in the main results.

W Auxiliary Loss Weight Ablation

Table 30: Effect of auxiliary persuasion loss weight λ during Stage-2 E2E fine-tuning (seed 42). Stage-1 DAPT encoder is identical across all rows; only the Stage-2 loss $L = L_{\text{binary}} + \lambda \cdot L_{\text{strategy}}$ varies.

λ	IID	DD OOD	FITD OOD	AA OOD
0	1.000	1.000	1.000	1.000
0.1	1.000	1.000	1.000	.990
0.5	1.000	1.000	1.000	.981
1.0	1.000	1.000	1.000	.981
2.0	1.000	1.000	1.000	.944

During Stage-2 E2E fine-tuning, the auxiliary persuasion loss weight λ has limited impact on detection (Table 30): the maximum AA OOD gap across all λ values is 5.6 pp (single seed), comparable to the cross-seed variance (∼6 pp) observed in the main results. All configurations converge to perfect validation F1 within epoch 1, indicating that DAPT representations already encode sufficient discriminative features for downstream classification (Section S). The value of persuasion strategy classification lies in Stage-1 DAPT—where it shapes encoder representations that transfer across attack families—not in the Stage-2 auxiliary loss.

X OOD Strategy Prediction Analysis

Table 31: OOD strategy prediction accuracy (jailbreak turns only) and detection rate for the 9-class model (3-seed mean). r: Pearson correlation between per-strategy prediction accuracy and per-strategy detection rate.

Dataset	Strat. Acc.	Det. Rate	Missed (per seed)	r
DD OOD	50.1%	99.6%	0 / 0 / 1	∼0.45
AA OOD	50.1%	96.2%	1 / 7 / 1	∼0.24

Table 31 reports OOD strategy prediction accuracy alongside detection rates. Strategy prediction

accuracy is substantially above random (50.1% vs. 11.1% chance for 9 classes) and positively correlated with detection success ($r \approx 0.24$–0.45), but strategy prediction is not the detection mechanism: frozen probing with a randomly initialized strategy head (Appendix S) achieves 97.2% AA OOD detection, and λ=0 (no strategy loss) matches or exceeds $\lambda > 0$ during E2E fine-tuning (Table 30). The classification objective shapes representations during Stage-1 DAPT training; at inference, strategy predictions are a byproduct of those representations, not a driver of detection.

Y Cross-Seed Ensemble

Table 32: Majority vote ensemble (3 seeds) vs. single-seed mean on OOD Full. "Fixed" = samples corrected by ensemble; "Broke" = samples corrupted by ensemble.

Variant	DD single→ens.	AA single→ens.	Fixed	Broke
9-class	.997→**1.000**	.972→**.981**	5	0
MLM	.914→**1.000**	.897→**.953**	26	0
Vanilla	.258→.258	.258→.258	0	0

Table 32 shows that majority vote never corrupts correct predictions (Broke = 0 for all variants). MLM benefits most from ensembling (+8.6 pp DD, +6.0 pp AA), because its per-seed errors are complementary—each seed fails on different samples, and the majority corrects these. This further exposes MLM's seed instability: its individual seeds are unreliable, but their errors are uncorrelated. By contrast, 9-class seeds are already highly consistent (std ≤ 0.026), leaving minimal room for ensemble improvement (+0.3–0.9 pp). Vanilla errors are systematic (all seeds fail on the same OOD samples), so ensembling provides no benefit.

Z Seed-Level F1 Comparisons

To verify that cross-seed pooling does not mask inconsistent pairwise orderings, Table 33 reports per-seed F1 for all variants on OOD DD and ActorAttack at full conversation length. The relative ordering of variants is consistent across all three seeds: classification-supervised variants (9-class, binary, scrambled) dominate MLM and vanilla at every seed, and no seed reverses the hierarchy established by the pooled analysis. Table 34 extends the comparison between 9-class and scrambled to 5 seeds for TOST analysis.

Table 33: Per-seed OOD F1 (Full) for all variants across three attack families (seeds 42, 123, 456). Cohen's d for 9-class vs. scrambled: DD = -0.003 (negligible), AA = 0.587 (medium; CI crosses 0 at n=3).

	DD			ActorAttack			FITD		
Variant	s42	s123	s456	s42	s123	s456	s42	s123	s456
9-class	1.000	1.000	.990	.990	.935	.990	1.000	1.000	1.000
Binary	1.000	1.000	1.000	.981	.990	1.000	1.000	1.000	1.000
Scrambled	.990	1.000	1.000	.891	.990	.971	1.000	1.000	1.000
MLM+cls	1.000	.990	.751	.971	.925	.784	.961	.432	.729
Topic	.727	.252	.644	.686	.431	.759	.990	.774	.909
Vanilla	.284	.305	.348	.258	.258	.258	.871	.855	.919

Benign recall = 1.000 for all DAPT variants across all seeds and OOD conditions (IID benign); OOD F1 variation is driven entirely by attack recall. Classification-supervised variants (9-class, binary, scrambled) achieve perfect or near-perfect F1 on FITD, confirming its non-discriminative nature.

Table 34: 5-seed OOD F1 (Full) for 9-class vs. scrambled (seeds 42, 123, 456, 789, 1000). At $n = 10$ (5 additional seeds; see text), TOST (δ=5 pp): DD $p < 10^{-8}$ ✓; FITD $p < 10^{-8}$ ✓; AA $p = 0.183$ (non-equivalent). Cohen's d at $n = 10$: DD ≈ 0; AA = 0.44 (medium).

Attack	**Variant**	s42	s123	s456	s789	s1000	Mean±SD
DD	9-class	1.000	1.000	.990	1.000	1.000	.998±.004
DD	Scrambled	.990	1.000	1.000	1.000	1.000	.998±.004
AA	9-class	.990	.935	.990	.972	.972	.972±.021
AA	Scrambled	.891	.990	.971	.990	.990	.966±.040